

Delayed Discounting Experiment
PSY310 - Lab in Psychology
Lab Report

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GitHub Link:

https://github.com/aryanm4/PSY310_LabInPsychology/tree/Delayed-Discounting

Introduction

Delayed discounting tasks quantify the propensity to discount the future rewards as compared to the present ones, which gives a behavior index of the impulsive decision-making.¹ It has been shown that there is always a significant variance in discount rates of people. The results of a meta-analysis by Amlung et al² revealed strong relationships between steeper delay discounting and psychiatric diseases such as bipolar disorder, borderline personality disorder, depression, and binge-eating disorder, where patients with such diseases demonstrated much stronger preference of immediate rewards over control groups. In addition, research on the subject of substance use disorders showed that the steepest discounts were observed in persons with alcohol, cocaine, nicotine, and opioid dependency compared to similar controls, indicating steep discounting as a transdiagnostic indicator of addiction severity. There are also cognitive and personality correlates with individual differences in delay discounting as the studies indicate that steeper discounting is correlated with lower intelligence, less inhibitory control and more self-reported impulsiveness, but the effect sizes in these studies are generally small to moderate.³ These results confirm that delay discounting tasks are useful in obtaining significant individual differences in impulsivity in a variety of populations and psychiatric disorders.

¹ Labvanced

² Amlung et al, 2016

³ Basile & Toplak, 2015

Method

Participants

The conducted experiment had 4 participants. They were all male undergraduates students at Ahmedabad University between the ages of 18 and 22. They were fully informed of the experiment's objective, to choose the more desirable reward, before it was conducted and full consent was provided.

Materials and Procedure

This experiment was created on a desktop computer with a resolution of 1920x1200 pixels using PsychoPy-2025.1.1. First, a cross shaped fixation was presented to the participant for one second, followed by the question. The question gave them a hypothetical reward of two Rupee amounts, one which they would receive today and the second which they would receive after a certain number of days. There were 100 questions, therefore 100 conditions and trials per participant. Participants were asked to click the 'T' key if they would like to receive the reward today, and the 'L' key if they want to receive the reward after the given number of days. The responses were collected by the PsychoPy software on an Excel spreadsheet, which were then analysed to determine the estimated discounting value.

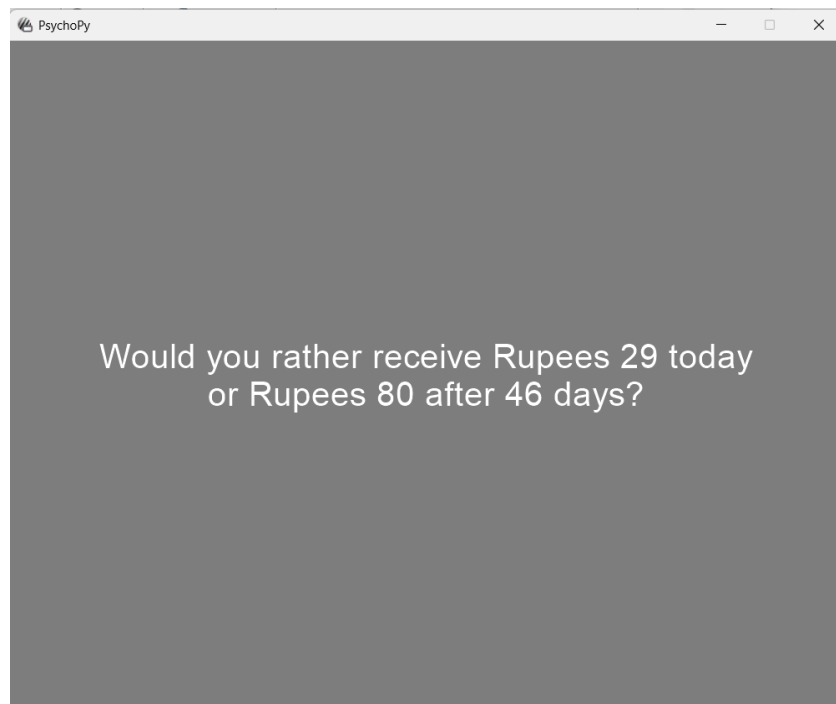


Figure 1 - Example of question provided to the participant

Results

Estimated discounting value:

Participant 1 - 0.00039851466

Participant 2 - 0.00241956141

Participant 3 - 0.00152604543

Participant 4 - 0.00257015366

Discussion

The k value represents how much a participant devalues a reward as the delay increases. Therefore, the results suggest that Participant 1 has the most patience in terms of receiving rewards, and would rather wait for a higher reward than receive an immediate lower reward. Participant 4, on the other hand, prefers immediate rewards even if they are lesser.

The discounting value is strongly recommended as a way of controlling individual differences in impulsivity and self-control as applied to behaviors in future. It uses people effectively in terms of preferences that lie between immediate and delayed rewards that are associated with significant psychological characteristics and behaviors. However, as much as discounting value is a strong and quantifiable indicator that has been widely applied in research, it has not been found to be perfect or exhaustive of the variations of individuals. The multi-dimensionality of impulsivity and decision-making is explained by the fact that the main characteristic of a discounted rate is time related to the preferences in choices. Individual differences are also brought about by other factors such as risk-taking ability, sensitivity to rewards and cognitive control. Besides, the rates of discounting may depend on the context, type of rewards and method of assessment.

References

Amlung, Michael, et al. "Steep Delay Discounting and Addictive Behavior: A Meta-Analysis of Continuous Associations." *Addiction*, no. 1, Wiley, Sept. 2016, pp. 51–62. Crossref, doi:10.1111/add.13535.

Basile, Alexandra G., and Maggie E. Toplak. "Four Converging Measures of Temporal Discounting and Their Relationships with Intelligence, Executive Functions, Thinking Dispositions, and Behavioral Outcomes." *Frontiers in Psychology*, Frontiers Media SA, June 2015. Crossref, doi:10.3389/fpsyg.2015.00728.

"Delay Discounting Task." Labvanced, <https://www.labvanced.com/content/research/en/blog/2025-06-delay-discounting-task/>. Accessed 4 Nov. 2025.